

AI-Enhanced Photo Authenticity: A User-Focused Approach to Detecting and Analyzing Manipulated Images

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Abstract—In the digital age, the authenticity of photographs is increasingly questioned due to the rise of sophisticated AI and human manipulation techniques. This paper presents the development of an AI-assisted system designed to aid users in identifying and verifying the genuineness of photos. Unlike fully automated solutions, our approach emphasizes user empowerment, providing tools to enhance human expertise in photo authentication. The system integrates advanced algorithms to detect potential manipulations and localize regions of interest, directing users' attention to specific areas that warrant closer examination. Users can conduct more efficient and effective authenticity checks without scrutinizing the entire image by focusing on these targeted regions. Our goal is to augment, not replace, the critical role of human judgment in verifying photo authenticity. The system's design and implementation are discussed, along with case studies demonstrating its practical application in various scenarios. This research aims to contribute to the growing field of digital forensics by providing a robust, user-friendly tool that bridges the gap between AI capabilities and human expertise in photo verification.

Index Terms—Photo Authenticity, AI-Assisted Verification, Manipulation Detection, Digital Forensics, Region Localization

I. INTRODUCTION

In the digital age and the widespread use of social media, images play a crucial role in information transmission and communication. The rapid development of digital image processing technology and computer vision has increased the number of digitally altered images. This has raised concerns about the authenticity of digital images, as it is now possible to create fake photos that are indistinguishable from real ones. Therefore, research on the capability to detect and localize fake images is highly urgent and significant.

Image forgery detection is the process of identifying and localizing altered images. This is a challenging task because there are various ways to manipulate images, and the techniques used for image forgery are continuously evolving. As the solutions for creating fake images become more sophisticated, developing tools to assist humans in making decisions

is essential. The approach we are interested in does not aim to replace humans in evaluating the authenticity of images but to develop solutions that support humans in analyzing and assessing forged images. This ensures accuracy and reliability in detecting and handling fake images, protecting the authenticity of visual information.

Our research aims to develop a flexible framework with easy-to-use functions to assist users in verifying the authenticity of photos. We are inspired by Trufor[7], a forensic framework for detecting image manipulation. Our proposed framework will be capable of integrating various modules to classify natural or AI-generated photos, detect deepfakes, and localize regions that may contain clues to identify AI or human manipulation. Additionally, it will incorporate visual search capabilities for retrieving similar photos from repositories, enabling users to compare and make informed conclusions about the authenticity of the images.

The main contributions in this paper are as follows:

- **Flexible Framework for Photo Verification:** Development of a versatile and user-friendly framework that integrates various modules to classify natural or AI-generated photos, detect deepfakes, and localize manipulated regions. This framework empowers users to verify the authenticity of photos efficiently.
- **Advanced Manipulation Detection and Localization:** Implementation of sophisticated methods to detect deepfakes and accurately localize regions within images that may indicate AI or human manipulation. This targeted approach enhances the precision and effectiveness of photo verification.
- **Visual Search for Comparative Analysis:** Introduction of visual search functionality that retrieves similar photos from repositories, enabling users to compare and analyze images. This feature supports informed decision-making in verifying the authenticity of digital images.

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Our proposed method involves a hierarchical application of noise-sensitive fingerprints to analyze high-resolution photos and detect subtle manipulations. We enhance the robustness of modification detection by using both RGB data and noise-sensitive fingerprints as inputs. We evaluate the effectiveness of our solution on real-world images, such as movie and advertisement posters, to verify its ability to accurately localize regions of interest, demonstrating its practical application and reliability. Besides, we also demonstrate our system to assist users in verifying photos (and associated text articles) based on retrieving similar images from photo and news repositories.

The content of this paper is organized as follows. In Section II, we briefly review existing work related to our task. Then, we present our proposed solution in Section III. Section IV discusses case studies and examples to illustrate our system. Conclusion and discussion are in Section V.

II. RELATED WORK

The issue of image forgery detection is on the rise. As digital image processing and computer vision technologies become more advanced, creating fake, indistinguishable images from real ones has become easier. This has raised significant concerns about the authenticity of digital photos, as they can be used to spread misinformation, commit fraud, and pose various other potential threats to individuals.

The issue of image forgery detection and localization has been extensively studied. There are two primary areas of research: forgery detection and forgery localization. Forgery Detection involves determining whether an image has been tampered with. It can be accomplished by identifying anomalies in the image that indicate forgery, such as inconsistencies in its metadata, compression artifacts, or unnatural image content. Forgery Localization involves identifying specific regions of an image that have been tampered with. It can be achieved using machine learning techniques to segment the image into different areas and then classify each region as either forged or genuine.

Different methods have been developed for forgery detection and localization. Statistical Methods look for anomalies in the statistical properties of the image that indicate forgery [10, 18]. Machine Learning Methods use machine learning algorithms to learn how to identify forged images [1, 3]. Feature-based methods extract features from the image that indicate forgery and then use these features to classify the image as forged or genuine [6, 9, 19]. Sharma et al. comprehensively evaluates various image forgery detection methods, ranging from traditional to deep learning approaches[14]. Their work discusses standardized datasets, benchmarks, evaluation criteria, and generalized strategies used in image tamper detection methods.

Recent research has focused on developing methods to detect and localize forgeries[7]. This is because forgery localization can provide valuable information about the nature of the forgery and the tools used to create the forged image. For example, Guillaro et al. propose TruFor[7], a forensic framework that detects and localizes various image

manipulations, including cheapfakes and deepfakes, using a transformer-based fusion of RGB images and a noise-sensitive fingerprint, offering pixel-level localization.

One challenge in forgery localization is distinguishing between natural and artificial image artifacts. This is because the same factors, such as noise, compression, and sensor errors, can cause both types of artifacts. Another challenge is the increasing sophistication of forgery techniques, making detecting and localizing forgeries more difficult. Forgers now use more sophisticated methods, such as deepfakes and GANs (Generative Adversarial Networks) [17].

III. PROPOSED SOLUTION

A. Overview

The architecture of our proposed framework consists of three main components: Non-genuine Photo Detection, Sensitive Region Localization, and Similar Photo Retrieval, as illustrated in Figure 1. Regular or expert users can use our system to check the authenticity of photos.

The **Non-genuine Photo Detection** component identifies whether a photo is genuine or tampered with by analyzing inconsistencies in metadata, compression artifacts, and unnatural content. The **Sensitive Region Localization** component pinpoints specific regions within an image that may have been altered, employing machine learning techniques to segment the image and classify each segment as either forged or genuine, directing users' attention to areas requiring further scrutiny. The **Similar Photo Retrieval** component aids in the verification process by retrieving visually similar photos from a repository, leveraging visual search algorithms to find comparable images, enabling users to compare and analyze differences, and assisting in concluding the authenticity of the photo in question.

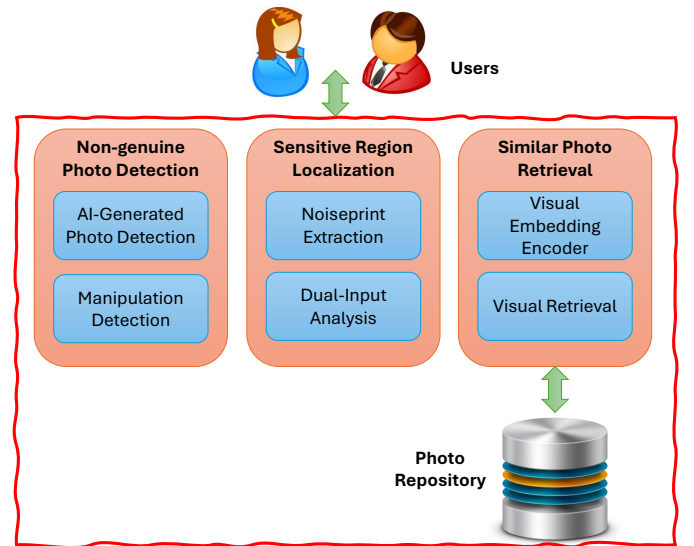


Fig. 1. Overview of our proposed system with three main components.

B. Non-genuine Photo Detection

The Non-genuine Photo Detection module comprises two key functionalities: AI-generated photo Detection and Manipulation Detection. AI-generated photo Detection identifies images created by artificial intelligence, distinguishing them from genuine photos. Manipulation Detection focuses on identifying tampered regions within an image, detecting inconsistencies and unnatural modifications that indicate forgery. We can integrate simple methods for AI-generated photo detection of manipulation detection, such as [12], [16, 5, 13].

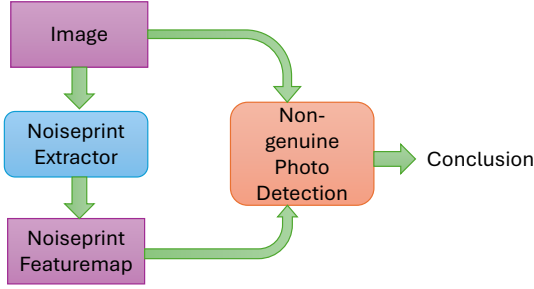


Fig. 2. Dual input for non-genuine photo detection.

Dual-Input Analysis: Our method utilizes both the RGB data and the noise-sensitive fingerprint (described in Section III-C) as two distinct inputs for assessing possible photo modifications. This dual-input strategy, shown in Figure 2 enhances the robustness of the detection process by leveraging different aspects of the image data, providing a comprehensive analysis of potential manipulations.

C. Sensitive Region Localization

The Sensitive Region Localization module includes two key components: Noiseprint Extraction and Dual-Input Analysis. Noiseprint Extraction generates a noise-sensitive fingerprint of the image to highlight areas that may have been tampered with. Dual-Input Analysis uses both the RGB data and the noiseprint [7] to comprehensively analyze the image, identifying specific regions that may contain signs of manipulation.

To effectively localize sensitive regions in an input image, we propose decomposing the image into multiple overlapping patches using different strategies, as illustrated in Figure 3. This hierarchical decomposition allows us to analyze the image at various granularities. For instance, in our experiments, an image is initially split into 3×3 patches, and this process can be recursively applied until the patch is nearly a uniform

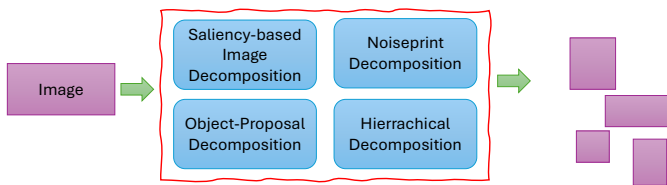


Fig. 3. Image decomposition for hierarchical analysis.

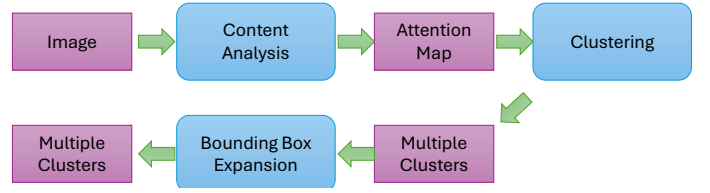


Fig. 4. Attention map-based clustering.

flat color. The hierarchical approach using noise-sensitive fingerprints helps us effectively handle high-resolution photos, enabling detailed and scalable analysis to detect subtle manipulations.

Given that large images can lead to high computational costs, we propose selecting only potential regions of interest for further analysis to save time and improve efficiency. This targeted approach helps maintain high accuracy while achieving better efficiency. Various strategies for image decomposition are employed, such as those based on saliency[2, 8], object proposals[15], or noiseprint maps[4]. For each approach, an attention map is created, such as a saliency map, object proposal map, or noiseprint map, to focus the analysis on the most relevant areas.

Figure 4 illustrates the process of localizing sensitive regions based on the attention map. The process of analyzing an image for sensitive region localization involves several steps. Initially, the image undergoes content analysis using techniques such as saliency detection[2, 8], object proposal methods[15], or noiseprint mapping[4]. Based on this analysis, an attention map is created to highlight areas likely to contain significant information or potential manipulations. The attention map is then used to cluster the main regions of interest within the image, and bounding boxes are drawn around these regions to define their extent. Each region is subsequently expanded by 150% in the area to capture the surrounding context, providing additional background information relevant to detecting anomalies. Finally, these expanded patches are analyzed to identify any anomalies, ensuring that even subtle manipulations within the context of each region are detected.

D. Similar Photo Retrieval

The Similar Photo Retrieval module consists of Visual Embedding Encoder and Visual Retrieval. The Visual Embedding Encoder converts the image into a compact, high-dimensional representation that captures its visual features. The Visual Retrieval component uses this encoded representation to search and explore a photo repository, retrieving visually similar images to aid users in comparing and verifying the authenticity of the photo in question.

The Similar Photo Retrieval module operates through a multi-step process. Figure 5 shows the main steps to assist a user in verifying the authenticity of a photo by looking for similar images in a repository or finding the possible sources that provide segments used in the photo of interest. Initially, the Visual Embedding Encoder processes the input image,

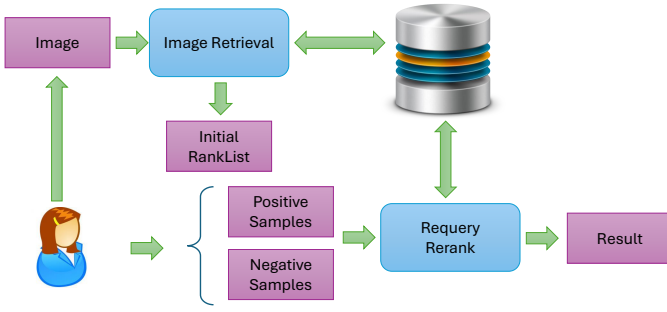


Fig. 5. Verification by users with iterative refinement of photo retrieval.

converting it into a compact, high-dimensional representation that captures its essential visual features. This encoded representation is used to initiate a search within the photo repository. The Visual Retrieval component then identifies and retrieves visually similar images. Users can review the ranked list of retrieved images and select several photos as positive or negative samples based on their similarity to the photo of interest. Using this feedback, the system performs a requery and reranks the candidate images, allowing for iterative refinement. This process enables users to gradually narrow down their search, finding images that are highly similar or that contain segments used to produce the photo of interest, thus providing a robust tool for verifying photo authenticity through visual similarity and iterative search refinement.

IV. EXPERIMENTS

A. Photo Analysis with Sensitive Region Hint

In this section, we demonstrate the usability of our system in assisting users to detect whether a photo is genuine or not through various case studies, including posters, advertisements, and photos generated entirely by AI.

The three cases—posters, advertisements, and photos generated entirely by AI—were chosen to demonstrate the versatility and robustness of our system across different types of images and contexts. Posters and advertisements often involve complex visual elements and potential manipulations, while AI-generated photos represent a modern and challenging type of forgery. These diverse case studies highlight the system’s capability to handle various scenarios, ensuring its practical applicability in real-world situations.

For movie posters, characters and objects of interest are usually combined into a single image, often involving copy-and-paste operations and various effects. The saliency map can detect different clusters, which, together with the noiseprint map, highlight various segments pasted from other sources to create the poster. The examples in Figure 6 illustrate the applicability of our approach in detecting these manipulations. In Figure 6, the left photo shows the movie poster, while the right photo displays the attention map highlighting sensitive regions for careful analysis.

For advertisements, key opinion leaders (KOLs) often provide their photos in poses appropriate for interacting with the

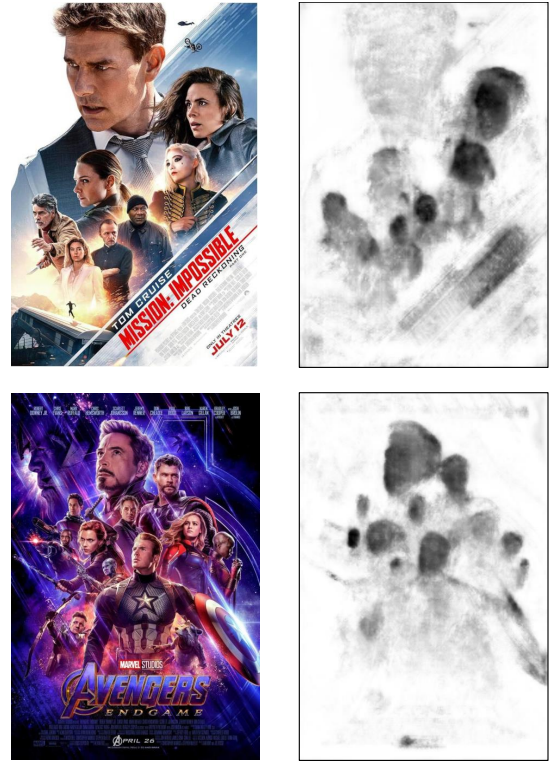


Fig. 6. Examples with movie posters.

advertised product. In the first row of Figure 7, we can easily recognize that the shampoo is highlighted in the hand of a famous player, suggesting that this object might have been pasted into the ad. In the second row of Figure 7, the attention map highlights the KOL’s photo, indicating that the person’s segment may have been placed into the ad’s background.

Figure 8 provides an example of a photo generated entirely by AI. Since the noise fingerprint in this photo is not from natural cameras, the attention map (based on the noiseprint) does not offer much useful information about sensitive regions for users to focus on. This highlights the need for more

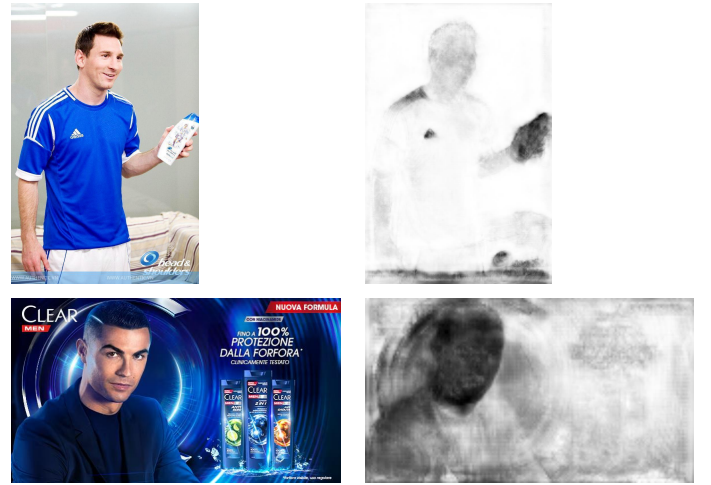


Fig. 7. Examples with advertisement photos.

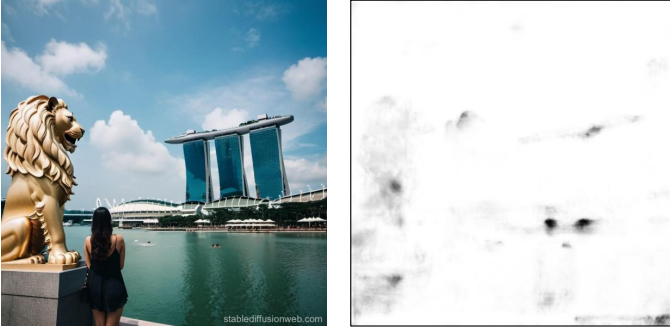


Fig. 8. Examples with AI-generated photo.

robust and appropriate methods to extract attention maps for AI-generated photos. The flexibility of our proposed system allows for the integration of new modules to address this challenge effectively.

B. Photo Verification with Retrieval Assistance

We illustrate how our system assists users in verifying the authenticity of photos by finding related photos in our repository. We consider a scenario where fake news is generated by synthesizing new photos from genuine ones. This experiment demonstrates the system’s effectiveness in detecting manipulated images and aiding users in making informed decisions.

We created a scenario where fake news photos are generated by manipulating genuine ones using Stable Diffusion with Analog Madness weight (base SD1.5).

Figure 9 shows two examples of generating fake photos (columns 1 and 2) from a genuine photo (column 3). To simulate a realistic scenario where fake news photos are generated by manipulating genuine ones, we performed the following steps. Genuine Photos: We started with 50 genuine photos of natural scenes (e.g., landscapes, building scenes). Using these genuine photos, we synthesized 100 fake photos depicting more disastrous situations using Stable Diffusion with Analog Madness weight (base SD1.5). We created a repository of 10,000 real photos from newspapers, including the 50 genuine photos used for fake image synthesis.

We invited 10 users to use our system to check the authenticity of 100 fake disastrous photos. Users inputted each fake photo into the system which then processed it using the Visual Embedding Encoder to generate a high-dimensional representation. The system searched the repository and retrieved

a ranked list of similar photos. This process significantly enhanced the users’ ability to determine authenticity. Users quickly retrieved the original (genuine) photo, usually among the top-ranked results, within less than 1 minute. With both the original and fake photos side by side, users could easily recognize the manipulations and suspect the authenticity of the more disastrous photos.

We also evaluated our system’s assistance for users in dealing with cheapfakes, where the same photo is used “out-of-context” in different articles. This scenario tests the system’s ability to help users identify the misuse of genuine photos by providing contextual information. We prepared a repository of 10,000 photos and related text articles. Some photos appeared in more than one article, potentially posing “out-of-context” situations if the related articles contained contradictory content, leading to potentially unreliable news.

10 users were requested to find potentially suspicious photos in out-of-context scenarios with 50 test cases, including 25 normal cases and 25 out-of-context ones. Users inputted a photo into the system which searched the repository for identical or similar images. The system returned users a ranked list of similar photos, including the news articles containing those photos. By comparing the input photo with retrieved results and reviewing the associated articles, users could recognize the misuse of photos in different articles, identifying potentially unreliable news.

Figure 10 demonstrates an example photo used in two articles with different contexts:

- Caption 1: “A health worker in Indonesia protects herself against infection as she prepares to treat a patient.” (URL: <https://news.un.org/en/story/2023/12/1145107>).
- Caption 2: “An online search of Indianapolis nursing jobs spotlights the ongoing wage-war fueled by the shortage.” (URL: <https://www.wfyi.org/news/articles/indiana-hospitals-brace-for-impending-doom-of-nurse-shortage>).

We first introduced this example in [11]. In this paper, we evaluate how our system aids users in identifying cheap fake scenarios by providing contextual information through advanced retrieval techniques. By comparing the content of the two articles, users can identify the photo being used out-of-context, thus highlighting a potentially cheap fake scenario.

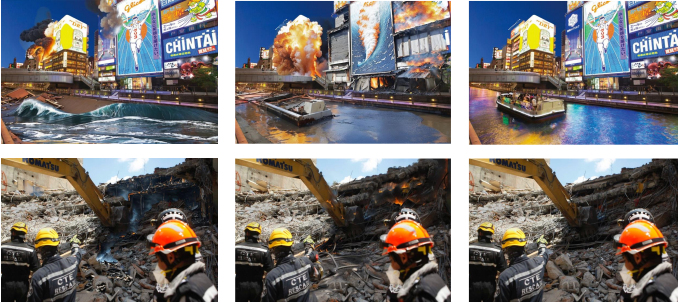


Fig. 9. Examples with generated fake photos from genuine ones.



Fig. 10. Example photo in two articles with potential out-of-context scenario.

V. CONCLUSION

We present a comprehensive framework for verifying photo authenticity using advanced detection, localization, and retrieval techniques. Our system integrates Non-genuine Photo Detection, Sensitive Region Localization, and Similar Photo Retrieval, effectively handling high-resolution images and identifying manipulated regions. The visual retrieval component allows users to iteratively refine searches and compare similar images from a repository.

Case studies on movie posters, advertisements, and AI-generated photos demonstrate the system's practicality and versatility. Users could accurately localize manipulated regions and verify image authenticity in various scenarios.

Experiments also showed the system's effectiveness in verifying news photos, including detecting synthesized fake photos and identifying out-of-context usage (cheapfakes). Users successfully retrieved original photos and identified misuse by comparing contexts.

Overall, our system provides a robust tool for detecting and analyzing manipulated images, enhancing the verification process through advanced retrieval techniques and user feedback.

Future enhancements will include integrating more sophisticated detection algorithms, expanding the repository, and incorporating natural language processing to analyze related articles for deeper context verification.

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